

HAT-P-37b

R-2401

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_190414 · created 2026-08-21T00:31:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458587.852 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.033
Transit depth (Rp/Rs)^2	1.6 +/- 0.82 [%]
Orbital Inclination (inc)	87.08 +/- 1.78
Airmass coefficient 1 (a1)	0.962 +/- 0.025
Airmass coefficient 2 (a2)	0.026 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	2.56 %
Best Comparison Star	#2 - [120, 215]
Optimal Aperture	3.04
Optimal Annulus	11.62
Transit Duration (day)	0.092 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.033 vs 0.1394 (deviation 0.41σ)
Duration fit vs published	0.092 d vs 0.0983 d (deviation 0.4σ)
Mid-time O–C	2.6 min
Depth SNR	3.8
Residual scatter	2.56 %

Light curve

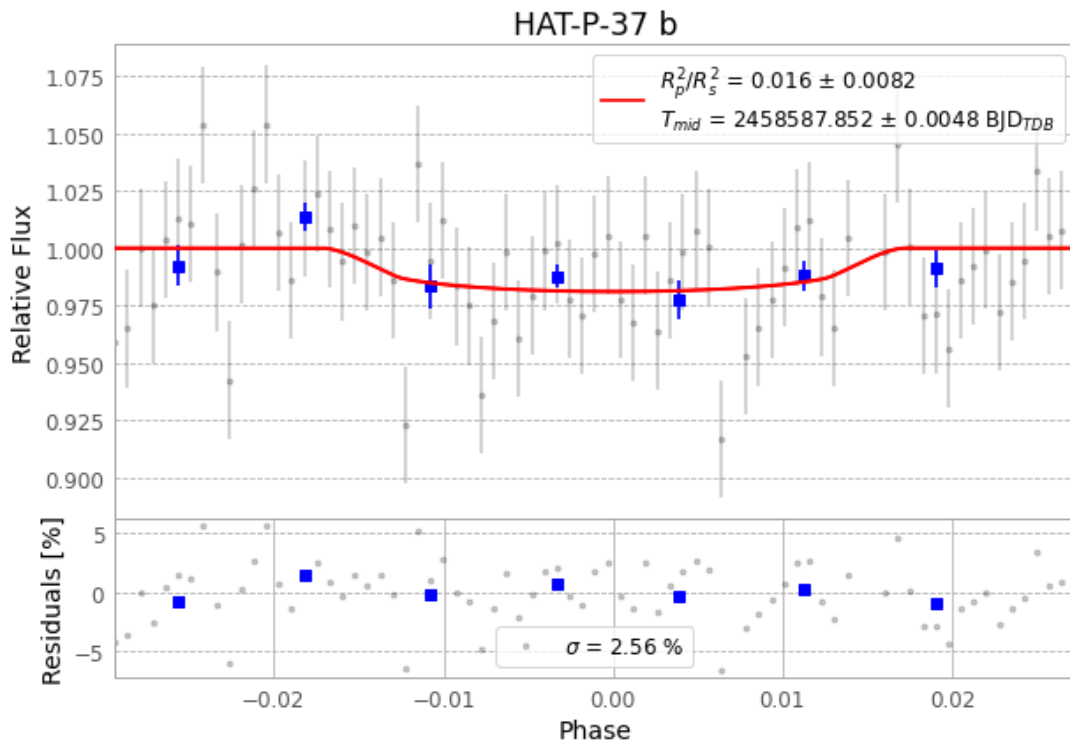


Figure 1 — FinalLightCurve_HAT-P-37 b_2019-04-13.png (SHA-256 1fdb36f36b8cc37b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 58], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 754c19a539c817d9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

77 FITS frames (51 MB), HATP-37190414062712.FITS → HATP-37190414101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-190414.log (9a3dfb2c1889...), FinalLightCurve_HAT-P-37 b_2019-04-13.png (1fdb36f36b8c...), FinalLightCurve_HAT-P-37 b_2019-04-13.csv (b3441e537f3a...)

Compute resources

Wall time 118.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:31Z.